CS215: Data Analysis and Interpretation

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Agenda

- Last time
  - Expectation of a function of a random variable
  - Leibnitz rule
  - Pareto principle
  - Course logistics
- Today
  - Linearity of expectations
  - Joint distributions

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Expected Value May Be Technically Undefined

$$P(X = x) = k/x^2 \text{ for } x \geq 1,\ x \in \mathbb{Z}^+$$

$$\sum_{x=1}^{\infty} P\,(X = x) = \sum_{x=1}^{\infty} k\,/x^2 = 1 \text{ if } k = 6/\pi^2$$

$$E(X) = \sum_{x=1}^{\infty} x\,P(X = x) = k \sum_{x=1}^{\infty} 1\,/x = \infty$$

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Pareto principle

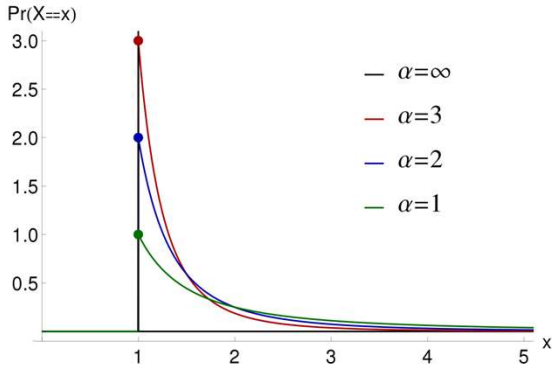
- The Pareto rule of thumb
  - 20% of pea pods contained 80% of the peas
  - 1992 United Nations Development Program Report
    - Richest 20% of the world's population receives 82.7% of the world's income
- A small fraction of causes often produces a large fraction of effects
  - Many things in life are not distributed evenly

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Pareto distribution

- Alpha = 1.16 exhibits the Pareto principle
- The type I Pareto distribution is characterized by a scale parameter x_m and a shape parameter α, which is known as the *tail index*.
- The expectation is technically undefined in some cases



$$f_X(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}} & x \geq x_m, \\ 0 & x < x_m. \end{cases}$$

$$\overline{F}(x) = \Pr(X > x) = \begin{cases} \left(\frac{x_m}{x}\right)^\alpha & x \geq x_m, \\ 1 & x < x_m, \end{cases}$$

$$\mathrm{E}(X) = \begin{cases} \infty & \alpha \leq 1, \\ \frac{\alpha x_m}{\alpha - 1} & \alpha > 1. \end{cases}$$

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Linearity of Expectations

- A function f is linear if f(ax) = af(x) and f(x+y) = f(x) + f(y)
- If f(x) = af(x) + b, then the function is affine, but commonly termed as linear
- Expectation is not a function but an operator — it takes a function as input — and it is linear in the same sense.

Typo → f(x) = ax+b

Aug 06 has correct form

$$E(a\,g(X) + b) = \int_{-\infty}^{+\infty} (a\,g(x) + b) f_X(x)\,dx$$

$$= \int_{-\infty}^{+\infty} a\ g(x) f_X(x)\,dx + \int_{-\infty}^{+\infty} b\ f_X(x)\,dx$$

$$= a\,E(g(X)) + b$$

Why is this useful?

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Mean squared prediction error

- You want to predict the value of a random variable X
  - Suppose we predict it as c
  - What is the error? Maybe $(X - c)$?
- Define the error to be the mean squared error
- Equality holds exactly when c = μ
  - The expected value is the predictor with least mean squared error.

Recall "sample mean"

$$E((X-c)^2) = E((X-\mu+\mu-c)^2)$$
$$= E((X-\mu)^2 + (\mu-c)^2 + 2(X-\mu)(\mu-c))$$
$$\geq E((X-\mu)^2)$$

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Absolute value

- What minimizes the quantity $J(c)$
  - The median
  - The median need not be unique

$$J(c) = \int_{-\infty}^{+\infty} |x-c| f_X(x)\, dx$$

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Variance

- Variance tells you how much the values of X deviates from the mean, on the average.
- The positive square root of the variance is the standard deviation
  - Low-variance densities concentrate around a point
  - High-variance densities are spread out

$$\mathrm{Var}(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x)\,dx$$

$$\mathrm{Var}(X) = E[(X-\mu)^2] = \sum_i (x_i-\mu)^2 P(X=x_i)$$

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Variance may be technically undefined

- For some distributions, the defining integral has no finite value, so the variance is undefined.
  - Example: the Pareto distribution with $\alpha < 2$.
- In some cases, the mean is defined but the variance is not
- In some cases, both are undefined.

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Hand Calculation of Variance

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2 + \mu^2 - 2X\mu] = E[X^2] + \mu^2 - 2E[X]\mu = E[X^2] + \mu^2 - 2\mu^2$$
$$= E[X^2] - \mu^2 = E[X^2] - (E[X])^2$$

- The middle step uses $E[X] = \mu$
  - This form — mean of the square minus square of the mean — is usually the easier one to compute with.
- Variance when you throw a fair dice
  - Earlier: expected value was 7/2
  - Option 1: $[(1-7/2)^2 + (2-7/2)^2 + \cdots (6-7/2)^2]\,(1/6)$
  - Option 2: $E[X^2] = 1/6(1+4+9+16+25+36) = 91/6$, Var = $91/6 - 49/4$

Recall "sample variance"

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Variance

- Shifting a random variable does not change its variance
- Scaling it scales the variance by the square.

$$\mathrm{Var}(aX+b) = E[(aX+b-E(aX+b))^2] = E[(aX+b-(a\mu+b))^2]$$
$$= E[a^2(X-\mu)^2] = a^2E[(X-\mu)^2] = a^2\mathrm{Var}(X)$$

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Jointly distributed random variables

- In machine learning one often needs to model relationships between two or more random variables.
  - CPI at IITB and the salary offered during placements
  - Hours spent on Instagram each day and hours of sleep at night.
  - Literacy level and crime rate
  - Price of a product, peer feedback and probability that a user clicks “Buy Now”
- If we observe multiple paired instances of X and Y, then some combinations of outcomes are more likely than others
  - Random variables rarely live alone. In real data, quantities interact, move together, or constrain one another.
  - Joint distributions captures in a mathematical way these relationships

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Joint Distribution function

- Recall: A random variable is synonymous with a cumulative distribution function, or simply a distribution function
  - We define the cumulative distribution function (cdf) $F_X(x) = \Pr\{X \leq x\}$
- The relationship between two random variables is defined by the joint cumulative probability distribution function

$$\mathrm{G}(x, \mathrm{y}) = \Pr\{X \leq x, \mathrm{Y} \leq \mathrm{y}\}$$

- All probability questions about X and Y can be answered in terms of G, for example

$$F_X(x) = \Pr\{X \leq x\} = \Pr\{X \leq x, Y < \infty\} = G(x, \infty)$$

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Joint pmfs

For discrete random variables *X* and *Y*, the **joint probability mass function** is

$$p_{X,Y}(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

The **pmf** of X follows by summing out the other

$$P\{X = x_i\} = P\left(\bigcup_j \{X = x_i,\ Y = y_j\}\right)$$

$$= \sum_j P\{X = x_i,\ Y = y_j\}$$

$$= \sum_j p(x_i, y_j)$$

This is called marginalizing

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three (and that's it)
  - Male and female children are equally likely and independent.
  - Let B (G) denote random variables for boys (girls)
  - What is the joint pmf p(B, G)
- No children? P(B = 0, G = 0) = 0.15
- One girl, no boy? P(B = 0, G = 1) = 0.2 × 0.5 = 0.1

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Example

- 15% of families are childless, 20% have one child, 35% have two and 30% have three. What is the joint pmf p(B, G)
- Two boys and one girl? P(B = 2, G = 1) = 0.3 × (1/8) × 3 = 0.1125
  - All 8 combinations of three children are equally likely; three of them have two boys and one girl.

$i \backslash j$	0	1	2	3	Row Sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column Sum $= P\{G = j\}$	.3750	.3875	.2000	.0375	

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